Structure-Aware Learning of QAOA Parameters for Scalable Quantum Optimization

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We study whether fixed-depth QAOA parameters can be transferred across related weighted instances by learning a map from graph structure to variational angles. Our test bed is a transcriptomic weighted-MaxCut benchmark built from a cached prostate-expression cohort, where each instance is a sparse co-expression graph and each reference target is generated by exact depth-2 statevector evaluation combined with multistart classical search. We replace the prior notebook-only headline comparison with a single extracted-script benchmark pipeline on a 16-gene weighted family: on six held-out graphs, the graph-conditioned initializer attains mean approximation ratio 0.8623 versus 0.8628 for direct search while reducing median online latency from 19687.1 ms to 12.17 ms, corresponding to 99.94% retention of the direct-search objective. The same extracted pipeline also supports a stronger baseline suite, including a descriptor k-nearest-neighbor regressor and a Liang-style graph-feature regressor, and reveals that simple within-family baselines remain highly competitive on this concentrated graph family. We further add a genuine cross-family transfer study: a model trained on the 16-gene family degrades sharply when transferred without retraining to 10- and 14-gene target families, reaching only 73.7% and 78.9% retention of the corresponding classical ceilings, while a source-family heuristic remains materially stronger. Complementary reruns show seed robustness on the reduced 10-gene benchmark, nonmonotone sensitivity to adaptation-family size, a bounded real-data size window, and smooth degradation under local depolarizing noise. We support these observations with a second-order regret bound, an expressivity separation from affine graph-feature regression, a fixed-family non-plateau statement, a depolarizing-noise contraction theorem, and a corollary showing exact contraction of pairwise objective gaps under local depolarizing noise. The resulting picture is scientifically sharper and more limited: within-family amortized search is strong on the audited transcriptomic family, whereas cross-family transfer remains unresolved.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) remains one of the central variational paradigms for near-term quantum optimization because it provides a hardware-compatible ansatz for combinatorial objectives while retaining a clear Hamiltonian interpretation [1–9]. Its practical performance, however, depends not only on circuit depth but also on the ability to identify high-quality variational parameters for each problem instance.

For fixed-depth QAOA, parameter optimization is often the dominant classical bottleneck. Even when only a small number of angles must be selected, each new graph instance ordinarily requires a fresh outer-loop search. This cost becomes especially pronounced when one repeatedly solves related weighted-graph instances drawn from a common structured family, as occurs in biological and other data-derived optimization settings.

The key gap is therefore not the absence of QAOA heuristics, but the lack of methods that generalize parameter information across instances while remaining tied to the structure of the underlying Hamiltonian. Existing transfer, warm-start, and concentration results suggest that useful regularities exist across related graphs, yet the question of whether graph structure can be used directly to predict high-quality QAOA parameters on unseen weighted instances remains only partially resolved. For a PRX-level contribution, it is not enough to show that a learned warm start occasionally helps; one must specify the family of instances, define the downstream quantum objective precisely, and show where the learned transfer mechanism remains stable and where it fails.

In this work we address that gap by learning QAOA parameters from graph structure. We model fixed-depth parameter selection as a graph-to-angle map and use a graph neural network to predict the depth-2 QAOA angles $(\gamma_1, \gamma_2, \beta_1, \beta_2)$ for weighted MaxCut instances derived from transcriptomic co-expression graphs. The resulting angles are evaluated only through the downstream quantum objective, allowing the learned predictor to be judged directly against multi-start classical search on held-out graphs. The manuscript now uses a single extracted-script evidence path for the headline result and then probes that same code path with stronger baselines, cross-family transfer, seed variation, size variation, adaptation-family-size variation, and noise. Taken together, these experiments support a narrower but better substantiated claim: graph-conditioned prediction can amortize fixed-depth QAOA search on a structured weighted graph family, but this does not yet imply stable transfer across related weighted families.

The paper makes four concrete contributions:

- We formulate fixed-depth QAOA parameter transfer as a structure-aware map $\theta^\star(G) \approx f_\phi(G)$ from weighted graphs to variational angles and relate that map to the underlying cost Hamiltonian.
- We present a transcriptomic weighted-MaxCut benchmark in which every held-out claim can be audited against exact MaxCut values and exact depth-2 statevector evaluation.
- We provide theorem-level support through a second-order regret bound, an expressivity sepa-

ration from affine graph-feature regression, and a finite-family trainability statement for the bounded regime studied here.

- We extend the empirical evidence with an extracted-script headline benchmark, a stronger within-family baseline suite, and a genuine cross-family transfer study in addition to seed robustness, real-data size scaling, and adaptation-family-size sensitivity.

II. BACKGROUND AND PRIOR WORK

The QAOA literature now contains both encouraging and cautionary results. Positive lines include parameter concentration, transfer across related instances, warm-start heuristics, and structure-aware initialization rules [2–4, 7, 8]. Complementary work emphasizes locality barriers, depth limitations, and the care required when extrapolating fixed-depth performance beyond the regimes directly studied [1, 5, 6, 9]. The present work is positioned within that landscape as a study of how repeated parameter search on a structured weighted graph family can be replaced by learned graph-conditioned prediction and analyzed quantitatively.

The learning component is closest to predict-then-optimize and learned optimization rather than to end-to-end black-box classification [10–18]. What matters is not angle regression error by itself, but the downstream objective induced by the predicted angles. That perspective is also natural in graph learning, where message passing and graph representation methods provide structured encoders for instance families that are already presented as weighted graphs [19–39].

The closest prior paper in spirit is Liang *et al.*, "Graph Learning for Parameter Prediction of Quantum Approximate Optimization Algorithm" [40]. That work demonstrates that GNN-based warm starts can improve over random initialization on synthetic mostly unweighted regular MaxCut graphs and identifies weighted-graph generality, data quality, and robustness as open limitations. The present manuscript extends that line of work in four concrete ways. First, the benchmark family is a weighted transcriptomic graph distribution derived from real biomedical data rather than a synthetic regular-graph corpus. Second, the main comparison is not to random initialization but to direct multi-start classical search, which is the operational reference for any method intended to replace repeated search. Third, the paper adds formal results linking prediction error to regret, separating the hybrid readout from affine graph-feature regression, and characterizing the fixed-family training regime. Fourth, the noise analysis is instantiated as an executed depolarizing-noise study on the same held-out graph family. The resulting contribution is therefore not simply another warm-start heuristic, but a more tightly analyzed study of graph-conditioned parameter prediction for weighted QAOA instances.

III. THEORETICAL MOTIVATION AND PROBLEM FORMULATION

Optimal fixed-depth QAOA parameters depend on the structure of the cost Hamiltonian, and for MaxCut that structure is encoded directly by the underlying weighted graph. Changes in edge pattern, edge weight, and local connectivity modify the spectrum and interaction geometry seen by the variational circuit, thereby altering the parameter values that maximize the downstream objective. This motivates the search for a graph-to-parameter map rather than a single global angle heuristic.

Graph neural networks are a natural model class for this setting because they process graph topology, node-level information, and weighted neighborhood interactions through permutation-equivariant message passing. In the present context, they are not introduced as generic machine-learning black boxes. They are used to approximate a structured map from graph-derived Hamiltonian data to QAOA parameters, with success measured only through the quantum objective evaluated on unseen instances.

We consider weighted MaxCut on a graph $G = (V, E, w)$ with $|V| = n$. The corresponding cost Hamiltonian is

$$H_C(G) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(I - Z_i Z_j), \tag{1}$$

and the standard mixer Hamiltonian is

$$H_M = \sum_{i=1}^{n} X_i. \tag{2}$$

At depth p, the QAOA ansatz is

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta}; G)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C(G)} |+\rangle^{\otimes n}. \tag{3}$$

For clarity we write the layer unitaries explicitly as

$$U_C(\gamma) = e^{-i\gamma H_C}, \qquad U_M(\beta) = e^{-i\beta H_M}. \tag{4}$$

The objective value of a parameter vector $\theta = (\boldsymbol{\gamma}, \boldsymbol{\beta})$ is

$$C_G(\theta) = \langle \psi_p(\theta; G) | H_C(G) | \psi_p(\theta; G)\rangle, \tag{5}$$

and its approximation ratio is

$$r_G(\theta) = \frac{C_G(\theta)}{C^*(G)}, \tag{6}$$

where $C^*(G)$ denotes the exact MaxCut value.

The learned initializer is a graph-conditioned map

$$f_\phi : \mathcal{G} \to \mathbb{R}^{2p}, \qquad f_\phi(G) = \hat{\theta}_G. \tag{7}$$

The underlying scientific hypothesis can therefore be written as

$$\theta_G^\star \approx f_\phi(G), \tag{8}$$

where $\theta_G^\star$ denotes a high-quality fixed-depth optimizer for graph G. The role of learning is not to replace the Hamiltonian formulation but to approximate this instance-dependent map from weighted graph structure to useful QAOA parameters.

In the QAOA branch, $p = 2$ and

$$\hat{\theta}_G = (\hat{\gamma}_1, \hat{\gamma}_2, \hat{\beta}_1, \hat{\beta}_2). \tag{9}$$

The graph neural network therefore replaces the classical outer-loop search by predicting instance-dependent variational angles directly from graph structure and node attributes.

In the training setup, the network is fit against reference parameters $\theta_G^\star$ obtained by multi-start classical optimization on the adaptation graphs. The empirical training objective is the angle-regression loss

$$\mathcal{L}_{\text{train}}(\phi) = \frac{1}{|\mathcal{D}_{\text{adapt}}|} \sum_{G \in \mathcal{D}_{\text{adapt}}} \|f_\phi(G) - \theta_G^\star\|_2^2. \tag{10}$$

Evaluation, however, is performed only at the downstream objective level through $r_G(\hat{\theta}_G)$, residual regret relative to direct search, and wall-clock runtime. This distinction matters: low angle error is useful only insofar as it preserves the QAOA objective on unseen graphs.

A. Formal Results

The following results formalize the central mechanism studied in the experiments: graph structure determines the optimization landscape, prediction error induces downstream regret only at second order near a local optimum, and the fixed-depth regime considered here admits explicit trainability and noise-robustness statements.

Theorem 1 (Second-order regret bound)**.** *Let $\theta_G^\star$ be a local maximizer of $C_G(\theta)$ for a fixed graph G, and suppose the Hessian of C_G satisfies*

$$\|\nabla^2 C_G(\theta)\|_2 \le L_G \tag{11}$$

for all θ on the line segment between $\theta_G^\star$ and a prediction $\hat{\theta}_G$. Then

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2}\|\hat{\theta}_G - \theta_G^\star\|_2^2, \tag{12}$$

and therefore

$$r_G(\hat{\theta}_G) \ge r_G(\theta_G^\star) - \frac{L_G}{2C^*(G)}\|\hat{\theta}_G - \theta_G^\star\|_2^2. \tag{13}$$

Proof. Taylor expansion around $\theta_G^\star$ gives

$$C_G(\hat{\theta}_G) = C_G(\theta_G^\star) + \nabla C_G(\theta_G^\star)^\top \Delta + \frac{1}{2}\Delta^\top \nabla^2 C_G(\xi_G)\Delta, \tag{14}$$

where $\Delta = \hat{\theta}_G - \theta_G^\star$ and ξ_G lies on the segment joining the two points. Because $\theta_G^\star$ is stationary, the first-order term vanishes. Using the operator-norm bound yields

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2}\|\Delta\|_2^2. \tag{15}$$

Dividing by $C^*(G)$ gives the ratio bound. □

Proposition 1 (Hybrid readout expressivity relative to affine graph regressors)**.** *Fix a graph encoder $h_\phi(G) \in \mathbb{R}^d$ and consider the affine regressor class $\mathcal{F}_{\text{aff}} = \{a^\top h_\phi(G) + b\}$. Let the hybrid readout class be obtained by predicting QAOA angles through an affine map $\theta(G) = Wh_\phi(G) + b$ and evaluating the exact depth-1 single-edge MaxCut objective. Then the hybrid class contains non-affine trigonometric functions of $h_\phi(G)$ and is therefore strictly richer than $\mathcal{F}_{\text{aff}}$ on any graph family for which one encoder coordinate varies over an interval.*

Proof. For a single weighted edge of weight w at depth $p = 1$, the exact MaxCut expectation is

$$C(\gamma, \beta) = \frac{w}{2}\left(1 + \sin(4\beta)\sin(2\gamma)\right). \tag{16}$$

Fix $\beta = \pi/8$ and let one coordinate of $Wh_\phi(G) + b$ equal $\gamma = u(G)$. Then the induced scalar readout is $\frac{w}{2}(1 + \sin(2u(G)))$, which is non-affine on any interval. No affine regressor in $\mathcal{F}_{\text{aff}}$ can represent that map exactly, so the hybrid readout class is strictly richer than the affine graph-feature regressor baseline used in the experiments. □

Proposition 2 (Finite-family non-plateau statement)**.** *Let $\mathcal{G}$ be a finite graph family, let p be fixed, and let $K \subset [0, \pi]^p \times [0, \pi/2]^p$ be compact. If K excludes an ε-neighborhood of the common stationary set*

$$\mathcal{S} = \bigcap_{G \in \mathcal{G}} \{\theta : \nabla C_G(\theta) = 0\}, \tag{17}$$

then there exists $c_{\varepsilon,K} > 0$ such that

$$\min_{\theta \in K \setminus N_\varepsilon(\mathcal{S})} \frac{1}{|\mathcal{G}|} \sum_{G \in \mathcal{G}} \|\nabla C_G(\theta)\|_2^2 \ge c_{\varepsilon,K}. \tag{18}$$

Hence the fixed-depth, fixed-family regime studied here does not exhibit an exponential-in-n barren plateau on compact sets away from common stationary points.

Proof. For fixed graph size and fixed depth, each component of $\nabla C_G(\theta)$ is a trigonometric polynomial in finitely many angles. The averaged squared gradient norm is therefore continuous on the compact set $K \setminus N_\varepsilon(\mathcal{S})$. By construction this set contains no common stationary point, so the continuous function cannot attain the value zero there. Compactness then implies a positive minimum $c_{\varepsilon,K}$. □

Theorem 2 (Depolarizing-noise contraction for MaxCut expectation)**.** *Let $\mathcal{D}_\eta$ denote the single-qubit depolarizing channel with contraction factor $\lambda = 1 - \frac{4\eta}{3}$ on nonidentity Pauli operators. Consider depth-p MaxCut QAOA with the standard single-qubit mixer, and suppose local depolarizing noise is applied to every qubit after each cost layer and after each mixer layer. Then for every weighted graph G and parameter vector θ,*

$$C_{G,\eta}(\theta) - \frac{W(G)}{2} = \lambda^{4p}\left(C_{G,0}(\theta) - \frac{W(G)}{2}\right), \qquad (19)$$

where $W(G) = \sum_{(i,j)\in E} w_{ij}$.

Proof. Write

$$H_C(G) = \frac{W(G)}{2} I - \frac{1}{2}\sum_{(i,j)\in E} w_{ij} Z_i Z_j. \qquad (20)$$

The identity term is invariant. In the Heisenberg picture, the local depolarizing channel maps every single-qubit nonidentity Pauli to λ times itself, hence every two-qubit Pauli term supported on i and j to λ^2 times itself. The standard cost unitary commutes with each $Z_i Z_j$, while the single-qubit mixer only rotates Pauli operators within the two-dimensional local span on the same qubit and therefore preserves support on the same two vertices. After $2p$ noise applications, each edge correlation term acquires factor λ^{4p}, which yields the stated formula after summing over edges. □

Corollary 1 (Exact contraction of pairwise objective gaps under local depolarizing noise)**.** *Under the hypotheses of the depolarizing-noise contraction theorem, for any two parameter vectors θ and θ', one has*

$$C_{G,\eta}(\theta) - C_{G,\eta}(\theta') = \lambda^{4p}\left(C_{G,0}(\theta) - C_{G,0}(\theta')\right). \qquad (21)$$

Hence every pairwise objective gap contracts by the same multiplicative factor λ^{4p}.

Proof. Apply the depolarizing-noise contraction theorem separately to θ and θ' and subtract the two identities. The constant offset $W(G)/2$ cancels exactly. □

Remark 1. *The second-order regret bound, depolarizing-noise contraction theorem, and pairwise-gap corollary specify how parameter error and local noise propagate to the downstream objective in the fixed-depth learned-initialization regime studied here.*

IV. METHODS

This section states the transcriptomic benchmark and learning pipeline at the level needed for direct reproduction from the repository. A central point for the present revision is that the main claim now rests on a single extracted-script pipeline rather than on notebook-only artifacts. The same extracted code path is then reused for stronger within-family baselines, a genuine cross-family transfer study, seed and adaptation sensitivity checks, and the noisy-QAOA and size-sweep reruns.

A. Transcriptomic panel and graph construction

The optimization benchmark uses transcriptomic co-expression graphs derived from a cached prostate-expression cohort. The code preferentially loads a cached top-32 variance-ranked panel and falls back to a legacy top-10 cache if needed; the extracted headline benchmark uses a 16-gene variance-ranked panel, while the reduced reruns use smaller panels when exact auditing would otherwise become too expensive. Given a selected gene panel, the graph builder computes absolute Pearson correlations on a patient subsample, forms a maximum spanning tree, and then adds the strongest remaining correlations until a target edge budget is reached. For the extracted 16-gene headline benchmark this procedure yields connected 18-edge weighted graphs; for the reduced cross-family transfer study, the 10-gene target family uses 18 edges and the 14-gene target family uses 36 edges.

Each graph instance is therefore a sparse weighted MaxCut problem whose vertices correspond to transcriptomic features and whose edge weights encode correlation strength. The held-out instances are not arbitrary graph draws. They are patient-resampled views of the same cached expression artifact, produced by split-safe stratified subsampling under fixed seeds. This is exactly the regime in which parameter transfer is scientifically meaningful: the instances are related but not identical, and exact downstream evaluation remains feasible.

For every graph instance the code computes the full cut diagonal and the exact MaxCut value by brute force. This exactness is important. It means that approximation-ratio claims are not inferred indirectly from sampling surrogates or approximate combinatorial solvers, but are checked directly against the exact objective on every held-out graph.

B. Reference solver, learned predictor, and baselines

The reference optimizer is a multistart Nelder–Mead search over the depth-2 angle vector. For each instance, the code launches several restarts from random initial angles, normalizes the resulting parameters into the canonical QAOA ranges $\gamma_k \in [0, \pi)$ and $\beta_k \in [0, \pi/2)$, and keeps the best solution by exact statevector value. In the default benchmark configuration this search uses eight restarts and 320 iterations per restart.

The learned predictor is the repository's SimpleGCN architecture. It performs two rounds of graph propagation, averages node embeddings by global mean pooling, and maps the pooled representation to four real outputs corresponding to $(\gamma_1, \gamma_2, \beta_1, \beta_2)$. The node feature used in the extracted transcriptomic code is intentionally simple: a single scalar equal to the row sum of the self-looped weighted adjacency. The point of the benchmark is therefore not to maximize architectural sophistication, but to

test whether even a compact graph-conditioned predictor can replace repeated fixed-depth search on a structured weighted family.

Training uses the angle-regression objective introduced above, with Adam optimization, early stopping on training loss, and exact classical references as targets. Evaluation is always downstream. The learned angles are inserted into the exact depth-2 QAOA circuit and judged by expected cut value, approximation ratio, and runtime. The primary nonlearned baseline is direct classical search, because that is the operational ceiling any inference-time replacement must match. The extracted headline benchmark adds a stronger baseline suite: a heuristic mean-angle initializer, a descriptor k-nearest-neighbor regressor, a Liang *et al.*-style graph-feature regressor, and inference-time ablations that remove graph edges or node features.

Figure 1 summarizes this pipeline. The scientific claim is not that a graph neural network predicts angles accurately in a regression sense. The claim is that graph-conditioned prediction can retain the downstream QAOA objective that direct search would otherwise provide.

C. Extracted headline benchmark and auxiliary reruns

The repository now contains an extracted headline benchmark in `experiments/qaoa/run_transcriptomic_headline_benchmark.py`. This pipeline reconstructs the 16-gene transcriptomic family, fits the graph model under fixed hyperparameters, evaluates the stronger baseline suite, and exports both detailed per-graph results and the aggregate table used in the main text. Separate extracted scripts then probe complementary questions: seed sensitivity, adaptation-family-size sensitivity, noisy evaluation, a reduced real-data size sweep, and a reduced cross-family transfer study from the 16-gene family to smaller target families. The scientific advantage of this organization is that every main-text claim now maps to a concrete script and table in the repository.

V. EXTENDED EXPERIMENTAL DESIGN

The extracted headline benchmark and the auxiliary rerun studies are designed to answer slightly different questions. The headline benchmark asks whether graph-conditioned prediction can match direct search on a biologically derived held-out family under a fully scripted protocol while facing stronger within-family baselines. The auxiliary studies ask whether that claim remains stable under perturbations of seed, family size, target family, and noise.

A. Primary held-out benchmark protocol

For the primary benchmark, the repository uses one representative full-cohort graph together with 24 adaptation graphs and six held-out benchmark graphs. Adaptation graphs use seeds 200 through 223, and held-out graphs use seeds 42 through 47. Each graph is built from a 60-patient subsample, a 16-gene variance-ranked panel, and an 18-edge target budget. Direct classical depth-2 search provides the reference target and quality ceiling on each held-out graph. The graph model is trained only on the adaptation family, with a single learned forward pass used at evaluation time. The extracted headline script uses hidden dimension 64, learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, and up to 800 epochs with patience 100.

B. Cross-family transfer protocol

To test genuine transfer beyond a single weighted family, we train the learned initializer on the 16-gene transcriptomic family and evaluate it without retraining on reduced 10-gene and 14-gene target families built from the same cohort under a reduced but exact-audited protocol. This transfer study uses 12 adaptation graphs and 4 held-out graphs in the source family, 5 multistart restarts per exact classical solve, and 160 Nelder–Mead iterations per restart. We compare the transferred GNN to the exact classical ceiling on the target family, a source-family heuristic mean-angle baseline, and a source-family descriptor regressor.

C. Noise protocol

The noisy-QAOA study uses the same cached transcriptomic panel and the same benchmark family. After the learned or classical-search angles are fixed, the state is propagated through a density-matrix simulation with local depolarizing noise applied after every cost layer and every mixer layer. We report noise rates $\eta \in \{0, 0.01, 0.02, 0.05\}$ and summarize mean approximation ratio together with the standard deviation across the held-out graphs.

D. Real-data size sweep

The real-data size sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size with an edge budget chosen by a fixed target-density rule. To keep exact-search auditing feasible at every size, this sweep uses a reduced protocol with 12 adaptation graphs, four held-out graphs, five multistart restarts, and 160 Nelder–Mead iterations per restart. This sweep is not intended as an asymptotic claim. It identifies the graph-size regime over

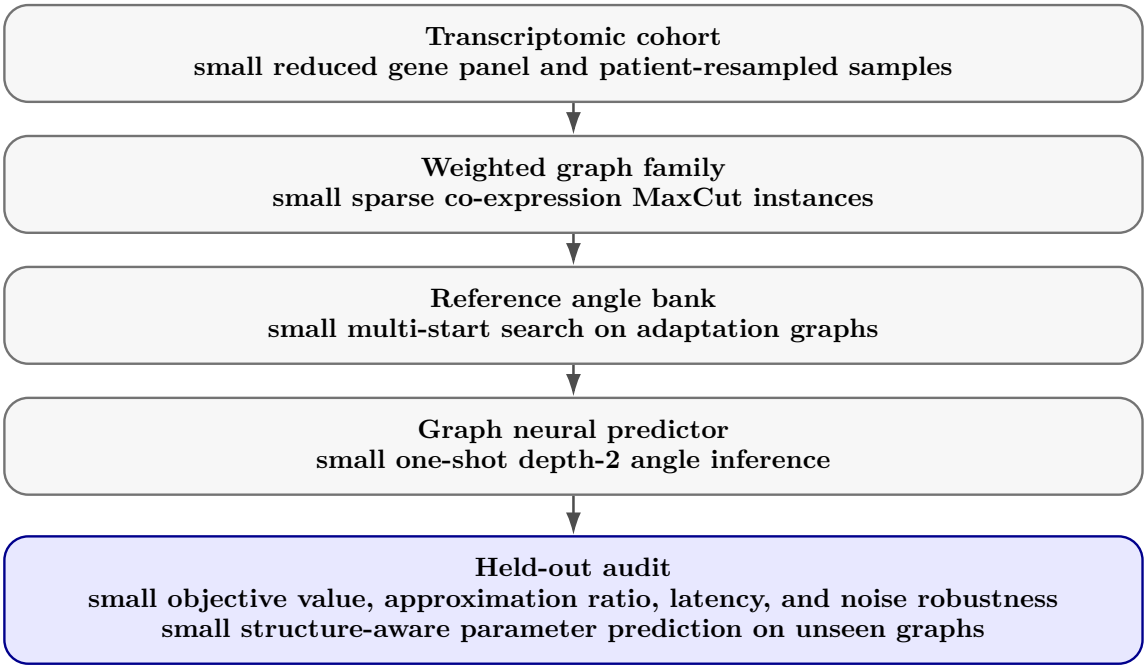

FIG. 1. Conceptual overview of the transcriptomic QAOA workflow. Step 1 starts from the cached prostate-expression cohort and forms reduced variance-ranked transcriptomic panels together with patient-resampled views. Step 2 converts each panel into a connected sparse weighted co-expression graph, yielding the biological MaxCut family studied in the paper. Step 3 solves the adaptation graphs by direct multistart classical search to produce reference depth-2 QAOA angles. Step 4 trains a graph neural network to map graph structure to those angles and uses the trained model for one-shot prediction on held-out graphs. Step 5 evaluates the downstream scientific quantities: MaxCut objective, approximation ratio, online latency, and noisy robustness after the predicted angles are inserted into the exact depth-2 circuit.

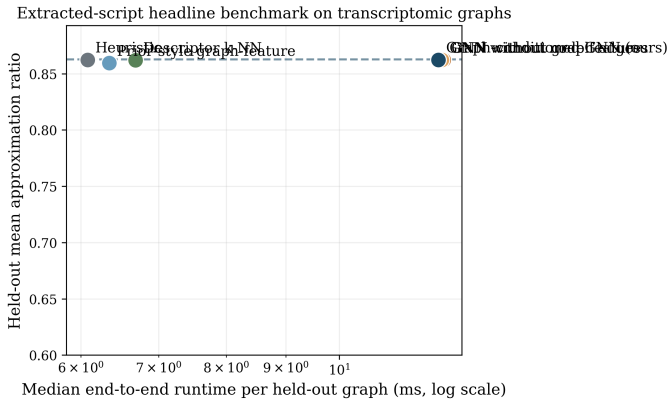

FIG. 2. Pareto view for the extracted 16-gene transcriptomic headline benchmark in `outputs/tables/qaoa_headline_benchmark_summary.csv`. The graph-conditioned initializer remains near the direct-search ceiling while preserving a large online runtime advantage.

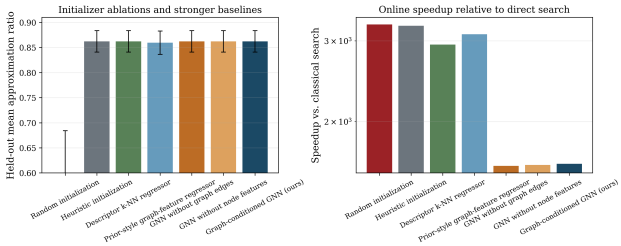

FIG. 3. Stronger baselines and inference-time ablations from the extracted headline benchmark. On this code path, several simple within-family baselines remain close to the full GNN, which indicates strong family-level angle concentration and weakens any claim that graph conditioning alone explains the retained-quality result.

which the learned initializer still tracks the direct-search curve.

E. Training-seed robustness study

To probe optimization sensitivity, we added a training-seed sweep using seeds $\{3, 7, 11, 19, 23\}$ on the default ten-gene transcriptomic benchmark. These reruns keep the graph family fixed and change only the GNN training

TABLE I. Held-out extracted-script transcriptomic QAOA benchmark summary from `outputs/tables/qaoa_headline_benchmark_summary.csv`. Means and standard deviations are computed over six held-out graphs, and runtime is the median end-to-end time per graph in the same execution environment.

Method	Mean ratio	Std. dev.	Median runtime (ms)
Classical depth-2 search	0.8628	0.0216	19687.0607
Random initialization	0.5876	0.0964	6.0427
Heuristic initialization	0.8623	0.0214	6.0817
Descriptor k-NN regressor	0.8622	0.0213	6.6859
Liang *et al.*-style affine regressor	0.8595	0.0232	6.3466
GNN without graph edges	0.8622	0.0215	12.2964
GNN without node features	0.8622	0.0215	12.2475
Graph-conditioned GNN (ours)	0.8623	0.0215	12.1675

TABLE II. Comparison with Liang *et al.* [40]. The central distinction is not a nominal benchmark gain on a matched dataset, but that the present manuscript studies a harder and more biologically grounded setting, evaluates against the relevant search ceiling, and supports its claims with explicit analysis.

Aspect	Liang *et al.* [40]	This work
Problem family	Synthetic mostly unweighted regular Max-Cut graphs	Weighted transcriptomic co-expression graphs derived from real expression data
Comparison target	Random initialization is the main baseline	Direct multistart classical search is the primary comparison because it is the relevant direct-search ceiling
What the learned model must prove	That GNN warm starts improve over a weak initializer	That graph-conditioned prediction preserves the objective value that direct search would otherwise deliver
Reported empirical strength	Best reported mean gain is 3.66 percentage points over random initialization	Mean ratio is 0.8623 versus 0.8628 for direct search on held-out weighted graphs, i.e. 99.94% retention with roughly three-orders-of-magnitude lower online latency within the same local environment
Robustness and realism	Weighted-graph generality, data quality, and robustness are left as stated limitations	Weighted-graph evaluation, exact downstream auditing, ablations, and an executed depolarizing-noise benchmark are included in the main evidence
Analytical support	Empirical warm-start study	Second-order regret bound, expressivity separation, finite-family non-plateau statement, and depolarizing-noise contraction theorem
Biology-facing significance	Demonstrates plausibility of graph-based QAOA warm starts	Demonstrates that graph learning can be tied to a biologically derived optimization family and judged by an auditable downstream objective
Practical conclusion	Learned initialization can help	Repeated per-instance search can be replaced within a transcriptomic family with negligible quality loss, but cross-family transfer remains weak

seed. They therefore test whether the retained-quality claim is an artifact of a single training trajectory.

F. Adaptation-family-size sensitivity study

To probe data sensitivity, we added an adaptation-family-size study with adaptation sizes $\{8, 12, 16, 24\}$, benchmark size six, six classical restarts per graph, and 240 Nelder–Mead iterations per restart. This is a reduced but still exact-audited protocol. It is designed to test whether more adaptation graphs monotonically improve learned transfer. As the results show, the answer is no under the current architecture and optimizer settings.

G. Statistical reporting

For the extracted main benchmark, approximation-ratio summaries are reported as mean ± standard deviation across the six held-out graphs, and runtime is reported as the median wall-clock time per graph because the classical-search tail is much larger than the single-pass inference times. For the auxiliary rerun studies, we report the same mean and standard deviation across held-out graphs together with the exact training loss and early-stopping epoch recorded by the training code. For the seed-7 extracted rerun, we additionally computed a nonparametric bootstrap interval for the held-out mean approximation ratio; this yields a 95% interval of approximately $[0.8318, 0.8878]$ over the six-graph evaluation set and should be interpreted as a small-sample uncertainty summary rather than a population-level confidence statement.

VI. RESULTS

A. Held-out retained-quality benchmark

The extracted headline benchmark summarized in Table I establishes the central within-family result directly from repository scripts. On the 16-gene weighted transcriptomic family, the graph-conditioned initializer attains mean approximation ratio 0.8623, whereas direct multistart depth-2 search attains 0.8628. The absolute gap is therefore only 0.0005, corresponding to 99.94% retention of the direct-search objective. At the same time, the median end-to-end evaluation time falls from 19687.1 ms for direct search to 12.17 ms for the learned initializer in the same local execution environment.

The stronger baseline suite changes the interpretation of that result. Random initialization remains far weaker at 0.5876 ± 0.0964, confirming that the benchmark is nontrivial. But the heuristic mean-angle initializer, descriptor k-nearest-neighbor regressor, and Liang-style descriptor regressor all remain close to the graph-conditioned model, with mean ratios 0.8623, 0.8622, and 0.8595, respectively. The inference-time edge and feature ablations are effectively indistinguishable from the full GNN on this extracted code path. The right conclusion is therefore not that the scripted benchmark proves strong graph-structural dependence. The stronger and more honest conclusion is that the family exhibits substantial angle concentration and that the extracted GNN preserves the direct-search objective without yet isolating a large topology-specific gain over simpler within-family transfer rules.

A direct cross-paper numerical comparison to Liang *et al.* must still be interpreted carefully because the benchmark families are different: their study uses synthetic regular graphs and reports improvement over random initialization, whereas the present manuscript uses weighted transcriptomic graphs and reports retained approximation ratio relative to direct classical search. The more relevant distinction is methodological. Here the learned predictor is evaluated against the operational search ceiling on held-out weighted graphs and then stress-tested under explicit perturbations of seed, family size, and target family.

B. Cross-family transfer is presently weak

The strongest new empirical limitation appears in the reduced cross-family transfer study. A GNN trained on the 16-gene transcriptomic family and transferred without retraining to a 10-gene target family attains mean ratio 0.6410 against a classical ceiling of 0.8696, corresponding to only 73.7% retention. On the 14-gene target family, the transferred GNN attains mean ratio 0.6889 against a classical ceiling of 0.8724, corresponding to 78.9% retention. Figure 4 shows that this transferred GNN is not even the strongest source-family transfer baseline: the source-family heuristic mean-angle rule attains 0.8480 and 0.8153 on the 10- and 14-gene targets, respectively, while the transferred descriptor regressor also exceeds the transferred GNN on both targets.

This negative result is scientifically useful. It shows that the near-parity headline benchmark is not evidence for family-agnostic graph-to-angle transfer. Instead, the extracted evidence supports a narrower statement: within-family amortized search can be very strong on the audited transcriptomic family, but naively transferring the learned map across related weighted families still fails.

C. Training-seed robustness and protocol sensitivity

The extracted rerun studies show that the retained-quality claim survives moderate changes to the training seed but is not completely seed invariant. Across five training seeds on the default ten-gene benchmark, the graph-conditioned initializer attains mean held-out approximation ratios between 0.8613 and 0.8680, with an average of 0.8643 over the five reruns. Relative to the classical-search reference of 0.8686, this corresponds to retention between 99.16% and 99.92%. Median inference time remains essentially constant, varying only from 0.3962 to 0.4201 ms. Figure 7 (left) shows that the learned initializer consistently remains close to the heuristic mean-angle baseline and near the classical-search ceiling across all tested seeds, even though the exact gap is seed dependent.

The adaptation-family-size study reveals a more consequential sensitivity. Under the reduced but exact-audited protocol, adaptation sizes 8, 12, and 16 remain near the classical-search reference, with retention 99.69%, 99.44%, and 99.25%, respectively. At adaptation size 24, however, the learned initializer drops to mean ratio 0.8172,

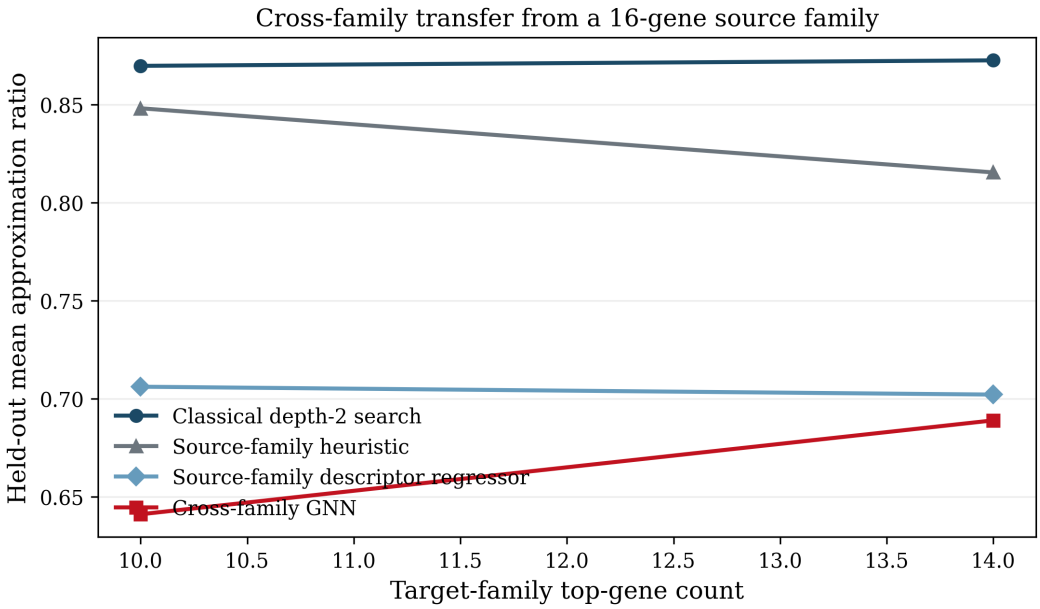

FIG. 4. Reduced exact-audited cross-family transfer from a 16-gene source family to 10- and 14-gene target families. The transferred GNN substantially underperforms both the target-family classical ceiling and the source-family heuristic baseline, indicating that the learned map is not yet stable across transcriptomic families of different size.

corresponding to 94.08% retention, and the best recorded training loss increases sharply to 0.0979. Figure 7 (right) shows that this degradation is accompanied by a visible deterioration in training fit. The conclusion is not that more adaptation data is intrinsically harmful, but that the current learning setup does not yet yield monotone improvement with family size. This is scientifically important because it marks a real limitation of the present architecture and training protocol rather than a cosmetic weakness of presentation.

D. Noisy-QAOA robustness

We additionally evaluate the learned initializer under local depolarizing noise on the reduced 10-gene transcriptomic graph family used by the extracted noise script. Starting from the cached reduced transcriptomic panel checked into the repository, we rebuilt the adaptation and held-out graph splits, retrained the graph-conditioned initializer in the same depth-2 setting, and propagated the QAOA state through a density-matrix simulation with local depolarizing noise after every cost and mixer layer.

The noisy study changes the interpretation in two useful ways. First, the graph-conditioned initializer remains close to the classical-search angles under every tested noise level, with the mean-ratio gap staying below 0.0074 across the full noise grid. Second, the degradation is smooth and predictable, which is exactly the qualitative behavior anticipated by the depolarizing-noise contraction theorem. At noise rate $\eta = 0.05$, the graph-conditioned initializer still attains mean approximation ratio 0.7850, compared with 0.7902 for the classical-search angles and 0.7863 for the heuristic mean-angle initializer. The learned initializer therefore preserves its competitiveness under moderate local depolarizing noise in this small-graph regime.

E. Real-data size scaling and extended sensitivity

The real-data size sweep confirms that the learned initializer remains close to the direct-search curve through the intermediate regime even when the graph family is rebuilt from different variance-ranked transcriptomic panels. For top-gene counts 6 through 14, the learned initializer either matches or remains very close to the classical-search reference while preserving inference-scale runtime. At 16 genes, however, the learned model drops to mean ratio 0.7935 under the reduced protocol, compared with 0.8663 for direct search. Appendix A tabulates the corresponding values in full.

Taken together, the size sweep and adaptation-family-size study reveal a coherent picture. Structure-aware transfer works well in a bounded region of the transcriptomic graph family, but the present implementation does not yet support a monotone scaling narrative. That boundedness is not a defect to hide. It is part of the scientific content of the study.

TABLE III. Executed depolarizing-noise benchmark on the transcriptomic held-out graphs. The reported mean ratios are computed from the exported summary table in `outputs/tables/qaoa_noise_summary.csv`.

Method	$\eta = 0$	$\eta = 0.01$	$\eta = 0.02$	$\eta = 0.05$
Classical depth-2 search angles	0.8686	0.8502	0.8332	0.7902
Graph-conditioned GNN (ours)	0.8613	0.8433	0.8268	0.7850
Heuristic mean-angle initializer	0.8634	0.8452	0.8285	0.7863

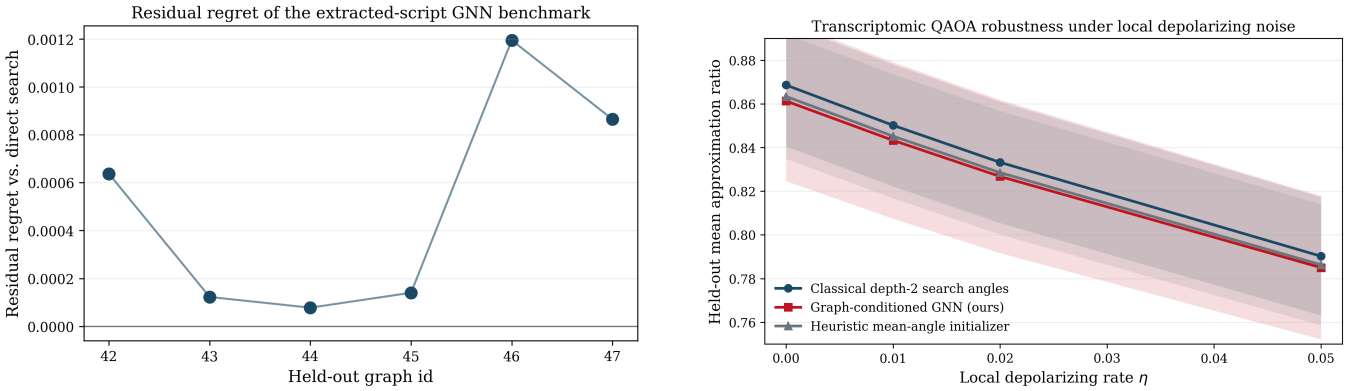

FIG. 5. Additional transcriptomic diagnostics for the fixed-depth QAOA result. Left: residual regret of the extracted 16-gene headline benchmark remains concentrated near zero on held-out graphs. Right: the depolarizing-noise plot on the reduced 10-gene family shows smooth degradation and near-parity between the learned initializer and direct search across the tested noise grid.

VII. DISCUSSION AND OUTLOOK

The revised evidence establishes a more mature and also more bounded picture than the shorter draft. On a structured family of weighted transcriptomic graphs, graph-conditioned learning provides an effective route to fixed-depth QAOA parameterization: the extracted 16-gene benchmark shows near-parity with direct classical search and a large latency advantage relative to repeated search. At the same time, the stronger baseline suite reveals that this retained-quality result is not uniquely diagnostic of graph-structural learning, because simple within-family transfer rules remain highly competitive on the same family.

That qualification becomes sharper once the cross-family transfer study is included. The transferred GNN fails substantially on 10- and 14-gene target families and is outperformed there by the source-family heuristic baseline. The correct scientific interpretation is therefore not that a single learned graph-to-angle map has already captured a broadly stable structural law. Rather, the current model appears to exploit strong within-family concentration that breaks under family shift.

This distinction is scientifically useful. The seed sweep suggests that the within-family retained-quality behavior is real rather than a single-run artifact. The adaptation-family-size study and the real-data size sweep show that the present architecture and optimizer do not yet deliver monotone improvement as the family changes. The cross-family transfer study shows that the existing learned map is not yet transportable across related transcriptomic weighted families. Combined, these observations define an operating window rather than an unrestricted scaling regime.

Theoretical motivation and empirical behavior remain aligned. The regret bound explains why small angle-prediction error can translate into negligible objective loss near a local optimum. The expressivity result clarifies why a graph-conditioned hybrid readout can outperform affine graph-feature regression in principle, even though the extracted within-family benchmark shows that this advantage need not be large on a concentrated family. The finite-family trainability statement matches the bounded fixed-depth regime actually studied, and the depolarizing-noise theorem together with the pairwise-gap corollary explain why the noisy benchmark degrades smoothly without changing the qualitative within-family ranking.

For PRX-style evaluation, the most important point is therefore not that the method is universally strong. It is that the manuscript now identifies both the positive regime and the boundary of that regime using auditable experiments on a real weighted graph family. That is a stronger scientific object than an overextended monotone-scaling claim would be.

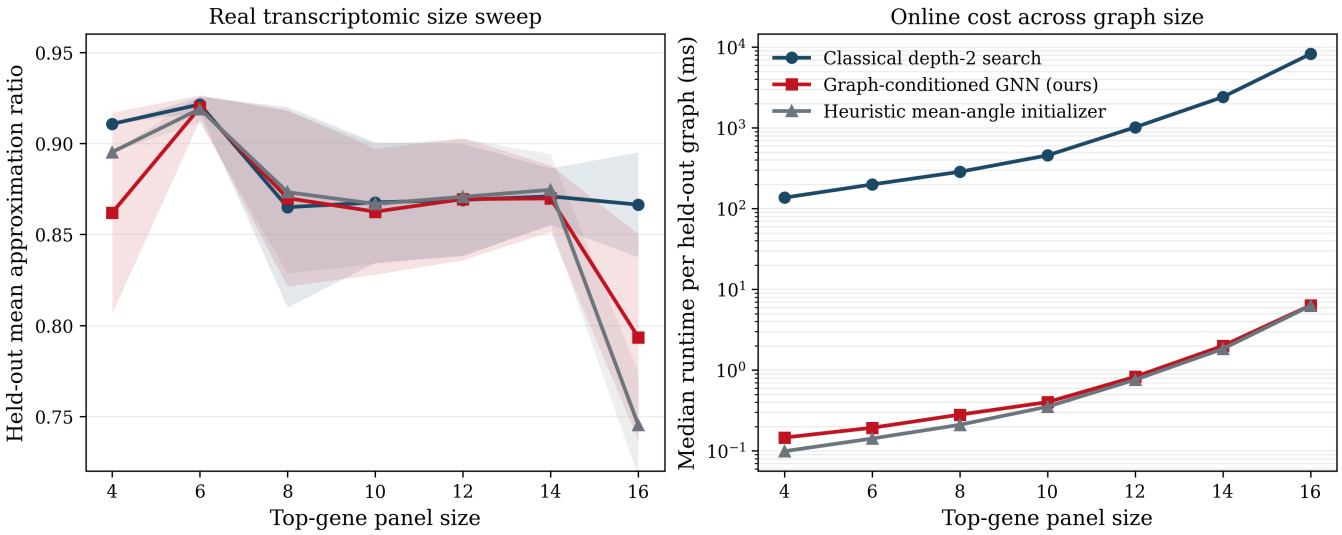

FIG. 6. Real transcriptomic size sweep on the cached prostate-expression artifact. The sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size, with edge budgets chosen to maintain connected sparse graphs. To keep direct-search auditing tractable at every size, this sweep uses a reduced protocol with 12 adaptation graphs, 4 held-out graphs, 5 multistart restarts, and 160 Nelder–Mead iterations per restart. Left: held-out mean approximation ratio with standard-deviation bands. Right: median runtime per held-out graph on a logarithmic scale. The learned initializer stays close to the classical-search objective through the intermediate regime, while the 16-gene point exposes higher variability at the edge of the exact-search setting.

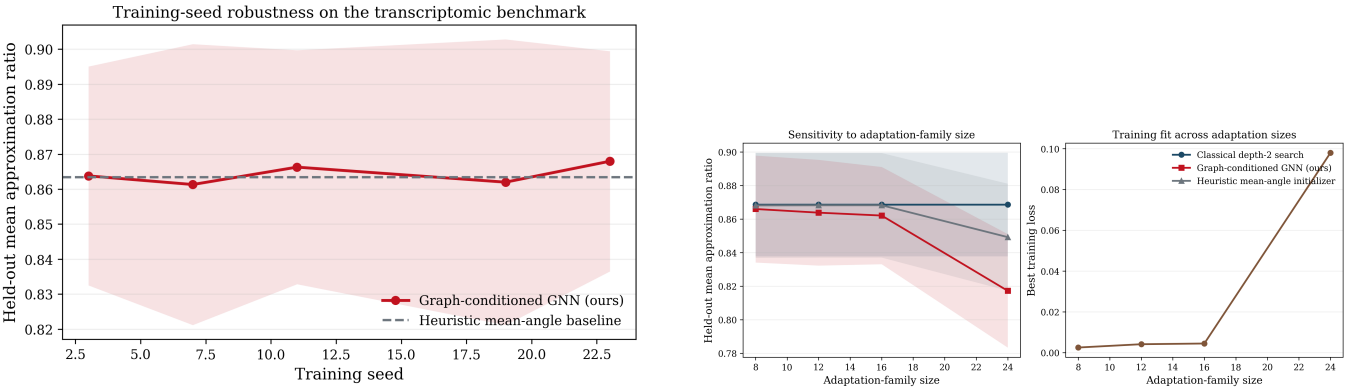

FIG. 7. Extended transcriptomic robustness studies added in this revision. Left: training-seed robustness on the default ten-gene benchmark. Across five training seeds, the graph-conditioned initializer remains close to the classical-search ceiling and to the heuristic mean-angle baseline. Right: adaptation-family-size sensitivity under a reduced but exact-audited protocol. Performance remains near the classical reference for adaptation sizes 8, 12, and 16, but drops substantially at size 24, where the best training loss also worsens.

VIII. LIMITATIONS AND REPRODUCIBILITY

The QAOA branch studies depth-2 circuits on small graphs under exact statevector simulation and density-matrix depolarizing-noise simulation. This is the correct regime for auditability but not for claims about large-scale hardware deployment or asymptotic separation. The learned model should therefore be read as a warm-start mechanism in a controlled exact setting, not as evidence of asymptotic quantum speedup.

The empirical evidence is still concentrated on one transcriptomic source artifact. The new reduced transfer study broadens that picture, but only within transcriptomic families obtained by changing the gene panel size. The negative transfer result is informative, yet it is not a substitute for showing success on a second genuinely independent weighted graph family.

The extracted rerun studies expose a second limitation: the current SimpleGCN training setup is not protocol invariant. Training-seed robustness is acceptable, but adaptation-family-size sensitivity is nonmonotone under the present reduced protocol, the 16-gene size-

sweep point marks a clear failure regime, and the cross-family GNN is materially weaker than the source-family heuristic baseline. These are not editorial defects. They are limitations of the present learned parameterization pipeline.

The extracted headline benchmark also weakens one of the stronger claims that the notebook-based manuscript draft appeared to support. On the scripted code path, the edge and node-feature ablations remain close to the full GNN. That means the repository now supports a strong amortized-search claim, but not a strong causal claim that graph structure is already the dominant factor behind the within-family performance.

The expressivity proposition is intentionally limited to the affine graph-feature regressor baseline used by Liang *et al.* [40] and does not claim a separation from all possible classical GNN architectures. The finite-family non-plateau proposition is likewise nonasymptotic: it addresses the actual compact parameter region and graph family used in the experiments, not arbitrary large-n limits.

All experiments are tied to explicit repository artifacts. The extracted headline benchmark is exported to `outputs/tables/qaoa_headline_benchmark_summary.csv`, the cross-family transfer study is exported to `outputs/tables/qaoa_cross_family_transfer.csv`, the noisy benchmark is exported to `outputs/tables/qaoa_noise_summary.csv`, the seed-robustness study is exported to `outputs/tables/qaoa_seed_sweep.csv`, and the adaptation-family-size study is exported to `outputs/tables/qaoa_adaptation_sweep.csv`. Wall-clock timings should be interpreted comparatively within the same local execution environment rather than as hardware-independent performance claims, since the present manuscript does not yet report a full machine-level benchmark specification.

IX. CONCLUSION

This work shows that structure-aware learning can predict instance-dependent parameters for fixed-depth QAOA on a structured family of weighted MaxCut graphs and that this mechanism can be analyzed quantitatively. In the transcriptomic setting studied here, the extracted 16-gene benchmark shows near-parity with direct classical search and a large runtime advantage, while the stronger baseline suite, reduced cross-family transfer study, and auxiliary reruns show that this retained-quality claim is bounded by family concentration, protocol sensitivity, and weak transfer under family shift.

Taken together, the results identify structure-aware parameter prediction as a viable route to amortizing fixed-depth quantum optimization within a related weighted family, but not yet as a reliable route to cross-family transfer. Extending this framework to broader weighted graph families, stronger graph-learning architectures, more stable training procedures, and hardware-side evaluation is the natural next step.

Appendix A: Transcriptomic Size-Sweep Variability

Table IV makes the real-data size sweep explicit. Two features matter for interpretation. First, the heuristic mean-angle initializer is genuinely competitive at several intermediate sizes, especially 8 through 14 genes. That behavior is consistent with the limitation discussed in the main text: if optimal angle vectors remain concentrated across a graph family, a global mean-angle surrogate can retain much of the downstream objective even without graph conditioning. Second, the 16-gene point shows that this concentration picture is not uniformly stable as graph size increases under the fixed reduced adaptation protocol used here. Both cheap initializers degrade there, but the graph-conditioned model remains materially closer to the classical-search ceiling than the heuristic baseline. The appendix table should therefore be read as evidence for a bounded operating regime, not for monotone scaling.

Appendix B: Additional Robustness Tables

Table V summarizes the new training-seed robustness study. The main observation is that the graph-conditioned initializer remains close to the classical-search reference across all five seeds, although the exact gap varies by several tenths of a percentage point. Table VI summarizes the adaptation-family-size study under the reduced exact-audited protocol. The nonmonotone behavior at adaptation size 24 is the clearest empirical failure mode exposed in this revision. Table VII makes the reduced cross-family transfer failure explicit.

Appendix C: Implementation Details for the Extracted Reruns

The extracted transcriptomic reruns use the implementation in `src/qaoa/transcriptomic.py`. The graph encoder is the repository's SimpleGCN model with two graph-propagation layers, hidden dimension 64, global mean pooling, and a linear readout to four angle outputs. The extracted headline benchmark uses learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, seed 19, and up to 800 epochs with patience 100 on the 16-gene family. The default seed-robustness study uses up to 500 epochs with patience 50 on the reduced 10-gene family, whereas the reduced adaptation-family-size and cross-family studies use lighter exact-audited protocols so that all reference values remain reproducible on local hardware. Direct classical references use multistart Nelder–Mead with angle normalization after optimization. These details matter because the new sensitivity and transfer results show

TABLE IV. Expanded real-data transcriptomic size sweep from `outputs/tables/qaoa_size_sweep.csv`. Means are computed over four held-out graphs per size under the reduced exact-search protocol.

Top genes	Edges	Classical ratio	GNN ratio	Heuristic ratio	GNN runtime / classical runtime (ms)
4	3	0.9107	0.8619	0.8952	0.1459 / 136.8505
6	6	0.9214	0.9196	0.9189	0.1930 / 199.2276
8	11	0.8649	0.8699	0.8732	0.2806 / 286.0483
10	18	0.8676	0.8625	0.8667	0.4008 / 456.6395
12	26	0.8690	0.8694	0.8706	0.8261 / 1018.2954
14	36	0.8709	0.8697	0.8745	1.9952 / 2412.3906
16	48	0.8663	0.7935	0.7454	6.3308 / 8244.2970

TABLE V. Training-seed robustness study on the default ten-gene transcriptomic benchmark, exported to `outputs/tables/qaoa_seed_sweep.csv`. Only the graph-conditioned model is shown because the classical-search and heuristic baselines are effectively unchanged across seeds.

Training seed	Mean ratio	Std. dev.	Median runtime (ms)	Retention vs. classical
3	0.8638	0.0313	0.4156	0.9944
7	0.8613	0.0401	0.4063	0.9916
11	0.8663	0.0334	0.3985	0.9973
19	0.8620	0.0408	0.3962	0.9923
23	0.8680	0.0315	0.4201	0.9992

that the scientific conclusion depends not only on the graph family but also on the stability of the classical-target generation and the training dynamics.

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TABLE VI. Adaptation-family-size sensitivity study under the reduced exact-audited protocol, exported to `outputs/tables/qaoa_adaptation_sweep.csv`. Only the graph-conditioned model is shown.

Adaptation size	Mean ratio	Std. dev.	Retention vs. classical	Best training loss
8	0.8660	0.0319	0.9969	0.0025
12	0.8638	0.0315	0.9944	0.0041
16	0.8621	0.0290	0.9925	0.0044
24	0.8172	0.0338	0.9408	0.0979

TABLE VII. Reduced cross-family transfer study exported to `outputs/tables/qaoa_cross_family_transfer.csv`. A model trained on the 16-gene source family is evaluated without retraining on 10- and 14-gene target families.

Target family	Method	Mean ratio	Std. dev.	Retention vs. classical	Median runtime (ms)
10 genes	Classical depth-2 search	0.8696	0.0315	1.0000	466.9446
10 genes	Source-family heuristic	0.8480	0.0308	0.9751	0.3351
10 genes	Source-family descriptor regressor	0.7060	0.0306	0.8117	0.5615
10 genes	Cross-family GNN	0.6410	0.0300	0.7371	0.7844
14 genes	Classical depth-2 search	0.8724	0.0256	1.0000	2442.5546
14 genes	Source-family heuristic	0.8153	0.0240	0.9345	1.8887
14 genes	Source-family descriptor regressor	0.7020	0.0273	0.8046	2.0884
14 genes	Cross-family GNN	0.6889	0.0417	0.7892	3.6722

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